LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Second-Hand Product Escrow Marketplace

Name	BR MANAVI	**SRN**	PES2UG24CS105
		Section	B
Course	Software Engineering	**Date**	9/9/2026

Architectural Justification

Architecture Selection

We chose Layered Architecture for the Second-Hand Product Escrow Marketplace. The system is divided into Presentation, Business, and Data layers so that user interaction, transaction logic, inspection/dispute processing, and persistent data management remain clearly separated.

Reason 1 - Clear Separation of Responsibilities

The marketplace has distinct responsibilities such as handling buyer/seller interactions, managing orders and escrow, performing independent inspection and verification, resolving disputes, and storing transaction records. Separating these responsibilities into layers makes the system easier to understand, develop, test, and maintain. Changes to the user interface or database can be made with less impact on the core business logic.

Reason 2 - Appropriate for the Transaction Workflow

The main transaction follows a structured sequence: purchase initiation, escrow locking, inspection, buyer verification, and final settlement. A layered design allows the business layer to coordinate this workflow while the presentation layer handles user actions and the data layer stores orders, escrow status, inspection reports, sign-offs, and disputes. This matches the defined workflow of the marketplace.

Security Advantage

Security-sensitive operations can be controlled in the business layer. Escrow release, dispute fund freezing, and verification checks are handled by the Order and Escrow Manager and Dispute Resolution Service rather than directly by the user interface. Inspection sign-offs and transaction records are maintained in the data layer, supporting controlled access and reducing unauthorized modification risk.

Performance Benefit

Separating the components allows frequently used functions such as order processing, inspection requests, and database access to be optimized independently. The business components can process transaction logic without unnecessary presentation-layer work, while data access is handled directly by the data layer. This supports responsive processing for time-sensitive escrow and inspection state changes.

Second-Hand Product Escrow Marketplace - Layered Component Architecture

